UNIT - II

HISTORICAL REVIEW

Quality Council - Statements - Strategic Planning - Deming - Barriers - Benefits - Leader - Gurus - Cases

1. Historical Review of Quality (background)

- **Pre-1900:** craftsmen made and inspected their own products; guilds set standards.
- **1900s - Taylor:** scientific management; separation of planning from doing; inspectors appointed.
- **1924 - Walter Shewhart** (Bell Labs): control chart, statistical quality control.
- **1940s - World War II:** military standards, sampling inspection (MIL-STD-105).
- **1950 - Deming and Juran** go to Japan (JUSE); Japanese industry adopts SQC and company-wide quality.
- **1960s - Zero Defects** (Crosby, Martin Co.); quality circles (Ishikawa, Japan 1962).
- **1970s - Japanese quality** (Toyota, TPS, Kaizen) captures world markets.
- **1980s - TQM** emerges in USA and Europe; 1987 Malcolm Baldrige National Quality Award; ISO 9000 series issued.
- **1990s onward** - Six Sigma (Motorola 1987, GE), QS 9000, ISO 14000, EFQM model.
- **2000s** - ISO 9001:2000 process approach; today ISO 9001:2015, Industry 4.0 and digital quality.

Core idea: quality moved from INSPECTION -> CONTROL ->

ASSURANCE -> TQM.

2. Quality Council

The **Quality Council** (also called Quality Steering Committee) is the group of senior managers who provide overall direction, guide and monitor the TQM effort of the organisation.

Composition

- Chairman: **Chief Executive Officer (CEO)** or top-most executive.
- Members: heads of production, marketing, finance, HR, design, purchase etc. and a representative of the union where relevant.
- The **Quality Officer / Quality Manager** acts as the co-ordinator (secretary).
- In large firms there are sub-councils at division and plant level.

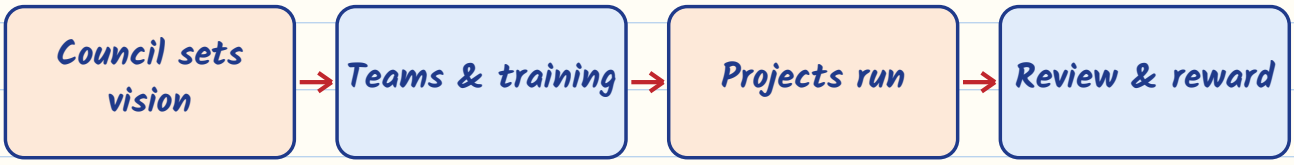
Duties and responsibilities

- Develop and review the **vision, mission and quality policy.**
- Estimate the cost of poor quality; set quality goals and strategy.
- Establish the TQM implementation plan - schedule, resources, budget.
- Set up and support **quality teams** (cross functional teams, quality circles).
- Review progress against goals regularly (monthly or quarterly).
- Approve training programmes; recognise and reward quality achievement.
- Remove barriers, communicate to all, and monitor customer satisfaction.
- Develop new measures when the old ones become inadequate.

Why a council is needed

- Shows visible top management commitment, gives continuity and coordination.

- Ensures that quality efforts in different departments do not conflict.



3. Quality Statements

Quality statements communicate the purpose and direction of the organisation. The three are **Vision, Mission and Quality Policy,** along with values/guiding principles.

a) Vision statement

> A short declaration of what the organisation wants to **become** in the future - its dream and long-term direction.

- Concise, inspiring, easy to remember, future oriented, customer centred.
- Eg. "To be the world's most trusted supplier of automotive components." / Eg. Microsoft: "A computer on every desk and in every home."

b) Mission statement

> The **purpose of existence** of the organisation - who the customers are, what is provided and how.

- Answers: Who are we? Who are our customers? What do we do? How do we do it better than others?
- Should be clear, brief (one or two sentences), specific, understood by all employees.
- Eg. "To provide reliable and affordable power products to meet or exceed our customers' expectations through continuous improvement."

c) Quality policy statement

> The overall intention and direction of an organisation related to quality as formally expressed by top management (ISO 9000).

- Must be **appropriate to the purpose** of the organisation; commit to meeting requirements and continual improvement of QMS.
- Provides a framework for setting quality objectives; communicated and understood at all levels; reviewed for suitability.
- Eg. Ford: "Quality is Job 1."
- Guidelines for writing: equal emphasis on customer, employee and supplier; short; the employees should be able to say it from memory; reflects top management's commitment.

d) Core values / guiding principles

- Customer first, integrity, respect for people, teamwork, continuous learning, social responsibility.

Statement	Question answered	Time frame
Vision	Where do we want to go?	Long term future
Mission	Why do we exist?	Present and ongoing
Quality policy	How do we commit to quality?	Operational, reviewed regularly

4. Strategic Planning

> **Strategic quality planning** is the process of establishing long-range (3 to 5 years) objectives and the strategies to achieve them, with quality built into the business plan - so that the customer needs are met and the organisation grows.

- Strategic planning is led by the **top management** and the quality council; it connects the vision to the day to day actions.

Seven steps of the strategic quality planning process (Besterfield)

1. **Customer needs:** find the needs and expectations through surveys, focus groups, complaints.
2. **Customer positioning:** where does the firm stand relative to competitors - benchmarking, SWOT.
3. **Predict the future:** technology, market, legal, economic trends.
4. **Gap analysis:** gap between the present position and the vision.
5. **Closing the gap:** choose strategies, develop the action plan.
6. **Alignment:** align the strategies with the mission, vision and values of the organisation.
7. **Implementation:** deploy objectives to departments, teams and individuals; allot resources.



Strategic planning terms

- **SWOT** - Strengths, Weaknesses, Opportunities, Threats.
- **Goals** - broad, long term; **Objectives** - specific, measurable, time bound (SMART).
- **Policy deployment (Hoshin Kanri):** the top level strategic goals are cascaded to every level; each manager draws his own objectives and means; reviewed periodically (catchball process).
- **Balanced scorecard** (Kaplan and Norton): financial, customer, internal process, learning and growth perspectives.

Strategic planning model - key inputs

- Mission, vision, customer data, competitor data, employee feedback, supplier input, past performance, regulatory requirements.

- Outputs: long term objectives, short term action plans, resource plan, performance measures.

5. Deming Philosophy

- **W. Edwards Deming** (1900-1993), American statistician, "father of the Japanese quality revolution". Deming Prize in Japan (1951).
- Philosophy: **quality is the responsibility of management;** 85% of problems arise from the system (management), only 15% from workers.
- Deming stressed reduction of **variation;** 'Chain reaction' - improve quality -> costs decrease (less rework) -> productivity up -> capture market -> stay in business -> provide more jobs.

Deming's 14 points for management

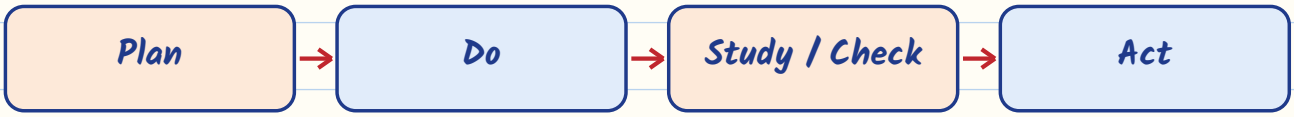
1. **Create constancy of purpose** for improvement of product and service.
2. **Adopt the new philosophy** - we are in a new economic age; do not accept defects.
3. **Cease dependence on inspection** to achieve quality; build quality in.
4. **End the practice of awarding business on price tag alone;** move to single supplier, long term relationship.
5. **Improve constantly and forever** the system of production and service.
6. **Institute training on the job.**
7. **Institute leadership** - help people do a better job; supervision, not policing.
8. **Drive out fear** so everyone works effectively for the company.
9. **Break down barriers between departments** - teamwork.
10. **Eliminate slogans, exhortations and targets** for the workforce (zero defects posters).

11. **Eliminate numerical quotas** and management by objectives; use leadership.
12. **Remove barriers that rob** workers of pride of workmanship; abolish annual rating/merit system.
13. **Institute a vigorous programme of education** and self-improvement.
14. **Put everybody to work** to accomplish the transformation - top management action.

Deming's seven deadly diseases

- Lack of constancy of purpose; emphasis on short term profits; evaluation by performance, merit rating or annual review; mobility of management (job hopping); running a company on visible figures only; excessive medical costs; excessive costs of warranty (fuelled by lawyers who work on contingency fee).

Deming cycle (PDCA/PDSA)



- Plan the change, Do it on a small scale, Study the results, Act - adopt, adapt or abandon. Repeat.

Deming's "System of Profound Knowledge"

- Appreciation for a system; knowledge of variation; theory of knowledge; psychology (human behaviour).

6. Barriers to TQM Implementation

- **Lack of management commitment** - quality treated as a slogan; no time and money given.
- **Inability to change organisational culture** - resistance to change, old habits, fear of job loss.
- **Improper planning** - no clear aims, no timeline, no sponsors.

- **Lack of continuous training and education;** wrong training at wrong time.
- **Incompatible organisational structure** and isolated departments - departmental silos.
- **Ineffective measurement techniques** and lack of data and results.
- **Inadequate use of empowerment and teamwork;** middle managers feel threatened.
- **Inadequate attention to internal and external customers.**
- **Poor communication** and lack of trust between management and employees.
- **Failure to recognise and reward** - no motivation.
- Short term thinking, expecting quick results, "flavour of the month" attitude.
- Paying inadequate attention to suppliers; copying other firms' programmes without adapting.

Overcoming the barriers

- Visible leadership, communication of vision, training at all levels, pilot projects with quick wins, involve unions, measure and publish progress, reward teams, review by quality council.

7. Benefits of TQM

Tangible benefits

- Improved product quality and reliability; fewer defects, rework and scrap.
- Lower cost and higher productivity; reduced cost of quality.
- Reduced customer complaints, warranty and returns.
- Increased sales, market share and profit; better delivery performance.
- Shorter cycle times and lower inventory.

Intangible benefits

- Higher customer satisfaction and loyalty; better company image.
- Improved employee morale, motivation and job satisfaction; teamwork and communication.
- Better supplier relationships; culture of continuous improvement.
- Better decision making with data; improved safety and environment.
- Competitive advantage; easier ISO certification, entry into global markets.

> Benefits flow: Quality up -> cost down -> productivity up -> customer satisfied -> profit up.

8. Characteristics of a Successful Quality Leader

- Gives **personal attention** to customers and quality; visible and active, "walks the talk".
- Has a clear **vision** and communicates it; sets the direction and creates the values.
- Builds **trust and respect;** honest, fair, ethical.
- Believes in **people:** delegates, empowers, trains, coaches and listens.
- Encourages **teamwork** and removes interdepartmental barriers; avoids blame - drives out fear.
- Fact based decisions; uses data and statistical thinking.
- Customer driven: knows customers, supplier partnership.
- Continuous learner, an agent of change, takes risks, accepts mistakes as learning.
- Recognises and rewards achievement.
- Long term thinking, patient, persistent; builds a quality culture; focus on prevention.

Leadership styles in TQM

Style	Feature
Autocratic	Leader decides alone - not suited to TQM
Participative	Involves team in decisions - preferred
Servant/transformational	Inspires change, supports people

9. Contributions of Gurus of TQM

1. W. Edwards Deming

- 14 points, PDCA cycle, 7 deadly diseases, variation (common vs special causes), Profound knowledge, statistical methods.

2. Joseph M. Juran

- "Fitness for use". **Juran Trilogy:** quality planning, quality control, quality improvement. Pareto principle (vital few, trivial many). Cost of quality. Book: Quality Control Handbook. 10 steps to quality improvement.

3. Philip B. Crosby

- "Quality is free". **Four absolutes:** quality = conformance to requirements; system = prevention; performance standard = zero defects; measure = price of non-conformance (PONC).
- **14 steps** to quality improvement. Quality vaccine. Quality maturity grid (uncertainty, awakening, enlightenment, wisdom, certainty).

4. Armand V. Feigenbaum

- **Total Quality Control** (TQC), "hidden plant", quality is everybody's work, the cost of quality, "quality at the source", 19 steps.

5. Kaoru Ishikawa

- **Cause and effect diagram** (fishbone), quality circles, 7 QC tools, company wide quality control (CWQC), internal customer "the next process is your customer".

6. Genichi Taguchi

- **Loss function,** robust design, design of experiments (orthogonal arrays), signal to noise ratio, off-line quality control.

7. Walter A. Shewhart

- Control charts, statistical process control, common and assignable cause, original PDCA.

8. Shigeo Shingo

- **Poka-yoke** (mistake proofing), SMED (single minute exchange of dies), zero quality control, just in time.

9. Taiichi Ohno and others

- Toyota production system, kanban, 7 wastes (muda). Masaaki Imai - **Kaizen.**

Guru	Main contribution
Deming	14 points, PDCA
Juran	Trilogy, fitness for use
Crosby	Zero defects, 14 steps
Feigenbaum	TQC
Ishikawa	Fishbone, QC circles
Taguchi	Loss function
Shingo	Poka-yoke

Juran vs Deming vs Crosby (comparison)

Point	Deming	Juran	Crosby
Definition	Predictable uniformity	Fitness for use	Conformance
Approach	Statistics, variation	Planning, control, improve	Zero defects
Responsibility	85% management	Management	Management
Standard	Continuous improvement	Cost of quality	Zero defects

10. Case Studies

Case 1: Toyota Production System

- Problem: low resources after WWII, need for high quality with low cost.
- Action: Just in time, jidoka (autonomation), kaizen, andon cord, quality circles, supplier partnership.
- Result: world leader in quality, very low defects, flexible production.
- Lesson: culture of continuous improvement and respect for people.

Case 2: Motorola (Six Sigma, 1987)

- Problem: field failures and high warranty cost; Japanese competition.
- Action: Six Sigma target of 3.4 defects per million, top management involvement, training of "black belts".
- Result: savings of billions of dollars, Baldrige award 1988.

Case 3: Xerox (Benchmarking)

- Problem: losing market share to Japanese copier makers in the 1980s.
- Action: benchmarking against Fuji Xerox, Canon and L.L.Bean; leadership through quality; employee teams.
- Result: improved quality and market share, Baldrige award 1989.

Case 4: Ford - "Quality is Job 1"

- Deming's advice adopted; employee involvement and supplier partnership; Taurus car success.

Case 5: Tata Steel / Indian examples

- Tata Steel won the Deming Prize (2008); TQM with Kaizen, quality circles. Infosys and Titan follow TQM and Six Sigma.

> How to write a case study answer: Background -> Problem -> TQM tools applied -> Results -> Lessons learned.

Important Exam Questions (Unit II)

Short answer

- What is a quality council? What are its duties?
- Differentiate vision and mission statements.
- List any five of Deming's 14 points.
- Name four barriers to TQM.
- Mention the four absolutes of quality by Crosby.

Long answer

1. Explain the strategic quality planning process.
2. Explain Deming's 14 points and the 7 deadly diseases.
3. Discuss the barriers and benefits of TQM implementation.
4. Describe the characteristics of a successful quality leader.
5. Discuss the contributions of Deming, Juran and Crosby.
6. Explain a case study on TQM implementation (Toyota / Motorola / Xerox).